



Figure 1: OpenSSH Protocol Stack

Table 1

Client authentication method	Description
Public authentication	Client and server have key pairs and exchange public keys during the authentication process
Password authentication	Plain text password for the given login user is used for authentication
Host-based authentication	Limits client access to a particular host/hosts
Keyboard authentication	Works on the basis of pre-stored security question and answer

People often mistake OpenSSH as a single utility installed on the server side, which is not true. In fact, it contains many small executables, each for a specific feature or purpose.

Table 2 lists such important tools from the protocol stack.

Table 2

sshd	Server daemon service
ssh	Client program to connect to daemon
sftp	Secure FTP for file transfer
scp	Secure file copy utility
ssh-keygen	Helps create keys for public key and host key-based authentication

Now let's look at the operations of each component as shown in Figure 1. The bottom layer that rides on the TCP protocol is the transport management layer, which is very important, because it takes care of crypto operations such as key generation and management. The next one is the user authentication layer, which is responsible for authenticating end users to the *sshd* daemon services, which are hosted on the server side. Once authentication is complete, a secure tunnel is created for data communication. This layer also ensures that you can use one of the many available ways of authentication such as the legacy username and password style, or the more advanced smart cards. The topmost layer is connection management, which adds features such as session multiplexing and also ensures additional security by blocking application layered attacks.

Installing OpenSSH

OpenSSH can be downloaded from openssh.com and the installation steps are as easy as for any other binary installation on a Linux distro. On Windows machines, running the installable exe file guides you through the steps. Remember to check if OpenSSH already exists on the machine; this is because many open source distributions come equipped with OpenSSH as a default SSH daemon. It is also important to secure the OpenSSH protocol, because, by default, it runs services in an unsecured fashion. Securing it involves studying and manipulating multiple parameters in the configuration files, which was covered in some articles in earlier issues of *OSFY*.

Features of OpenSSH

While there are many features available in OpenSSH, let's discuss those that are important from the security and operations point of view. First and foremost, it supports almost all operating systems available, right from the legacy *openbsd* project to the most recent Linux or Windows distribution. As mentioned earlier, it is an open source protocol stack and available under free licensing. OpenSSH has strong authentication capabilities. This is achieved by various levels of encryption as well as a few built-in features that block security holes, especially in the areas of routing and DNS spoofing. It also provides very strong encryption standards such as 3DES, AES, etc, which make it a perfect fit for industry standard installations. It offers protocol privacy, by which all authentication happens seamlessly and in a manner that is transparent to the user and systems, but at the same time, is a robust and verbose logging system that ensures audit trails of each activity. The way security is designed in the protocol is as if it never trusts the network or the remote connection.

As for the utilities, SFTP and SCP help copy files securely over the SSH tunnel between clients, or the client and server. OpenSSH also supports data compression which reduces chatter on the network, thus making more bandwidth available to other nodes. It also supports protocol tunnelling, whereby legacy protocols such as Telnet or *ftp* can piggy-back over an existing authenticated *ssh* tunnel. For advanced IT implementation, most secure protocols such as Kerberos and AFS ticket passing are also supported.

Summary

While OpenSSH is a versatile, robust and secure protocol, it is imperative to tune it up in order to ensure security. OpenSSH is available on all Linux distros as well as various versions of Windows. Read up on the documentation to understand the right and secure way of installing and configuring the protocol. **END** 

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